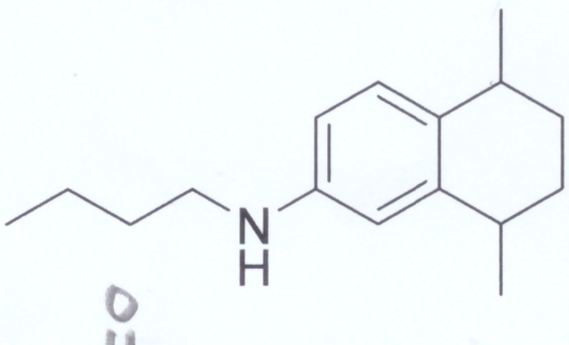
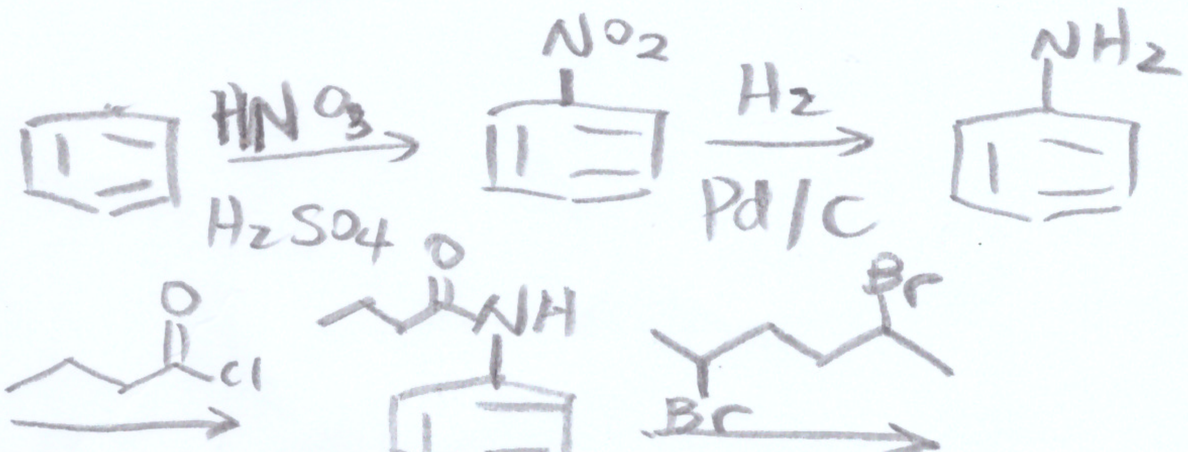
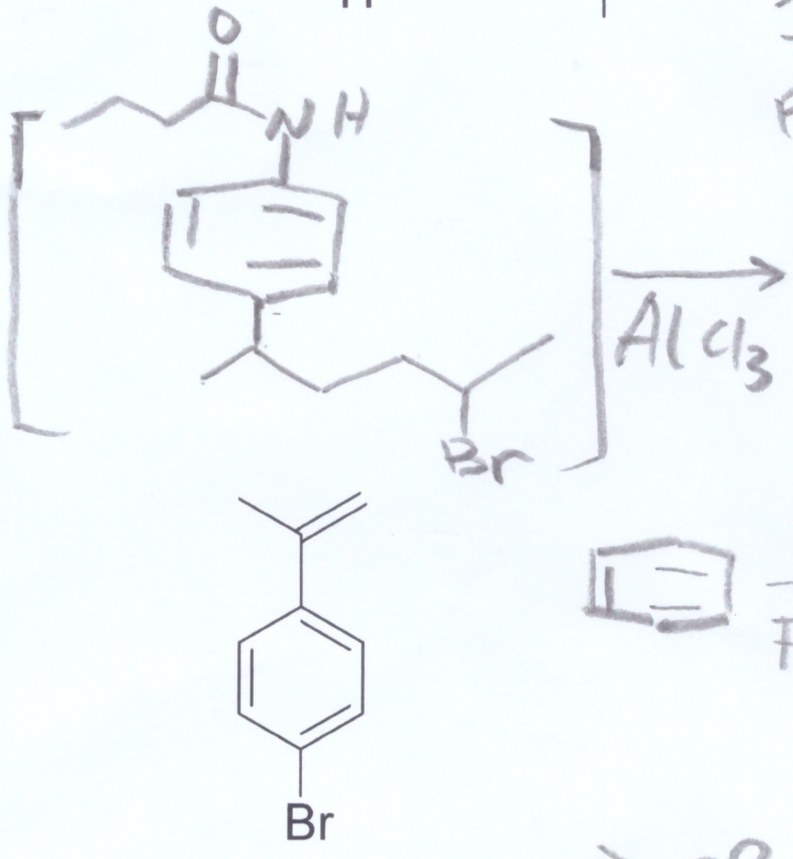
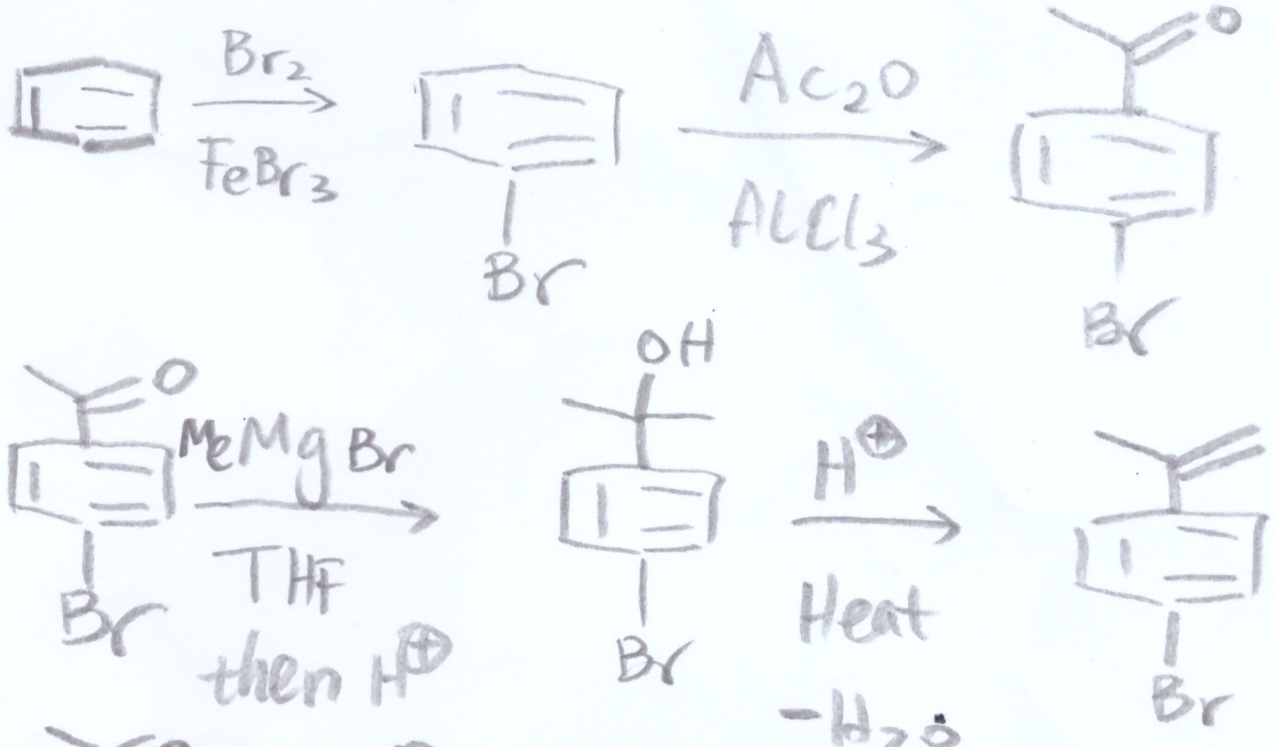
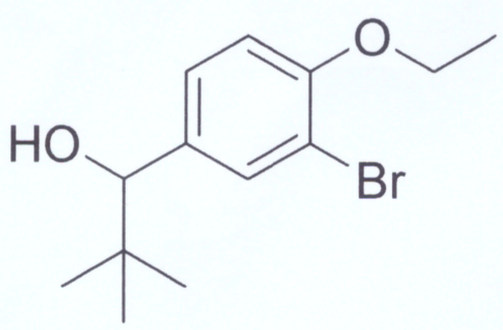
1. Propose concise synthetic routes for each of the following compounds, starting with molecules that contain no more than six carbon atoms.

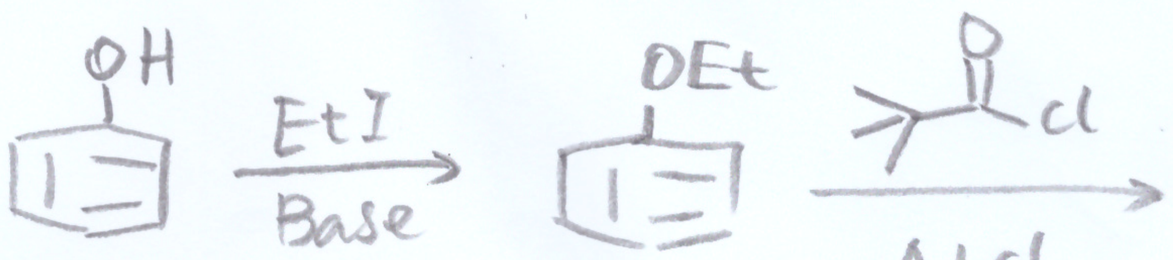


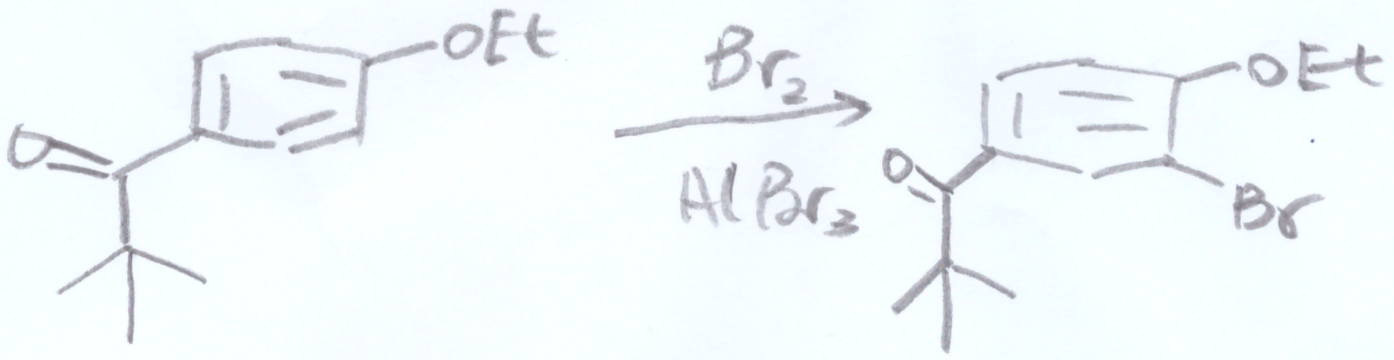




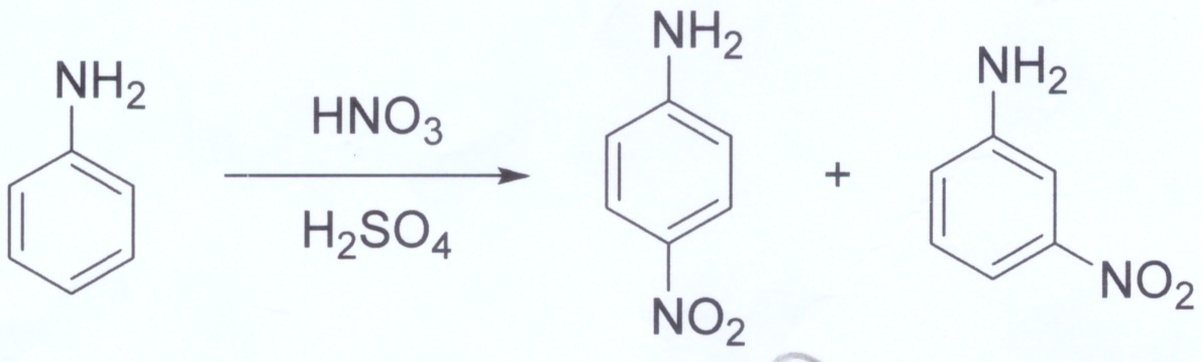


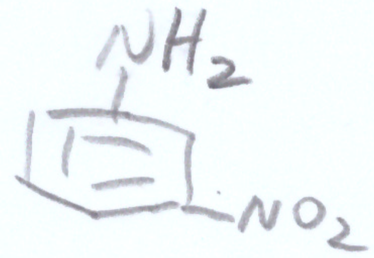




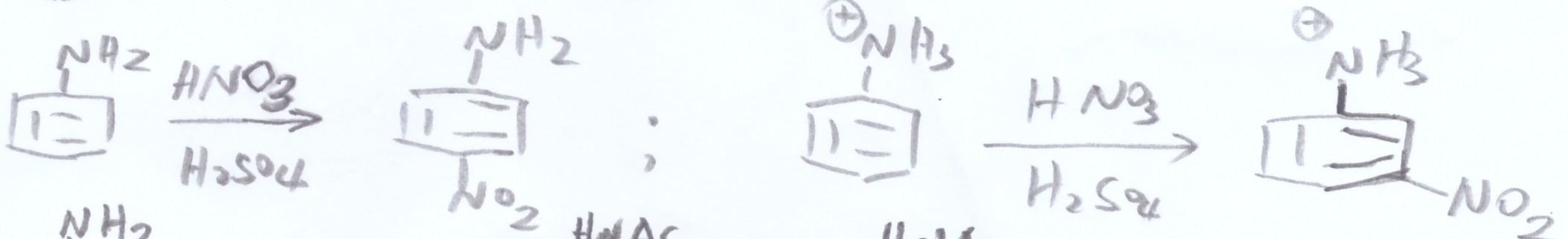


2. Nitration of aniline give a mixture of para-nitro and meta-nitro products (1:1 ratio).
   (1) Explain why both products are formed.
   (2) Propose a synthetic route from aniline to para-nitro aniline without the formation of meta-nitro product.

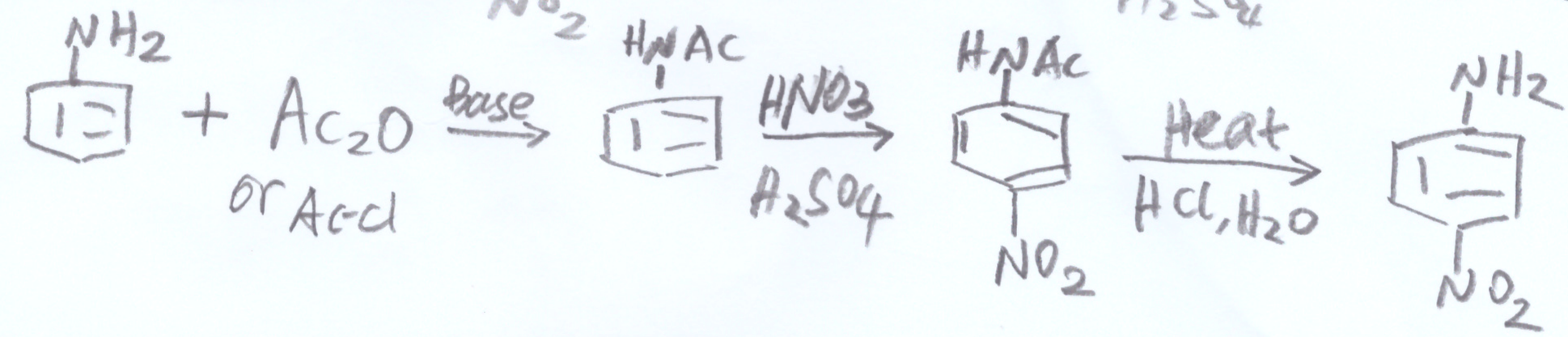




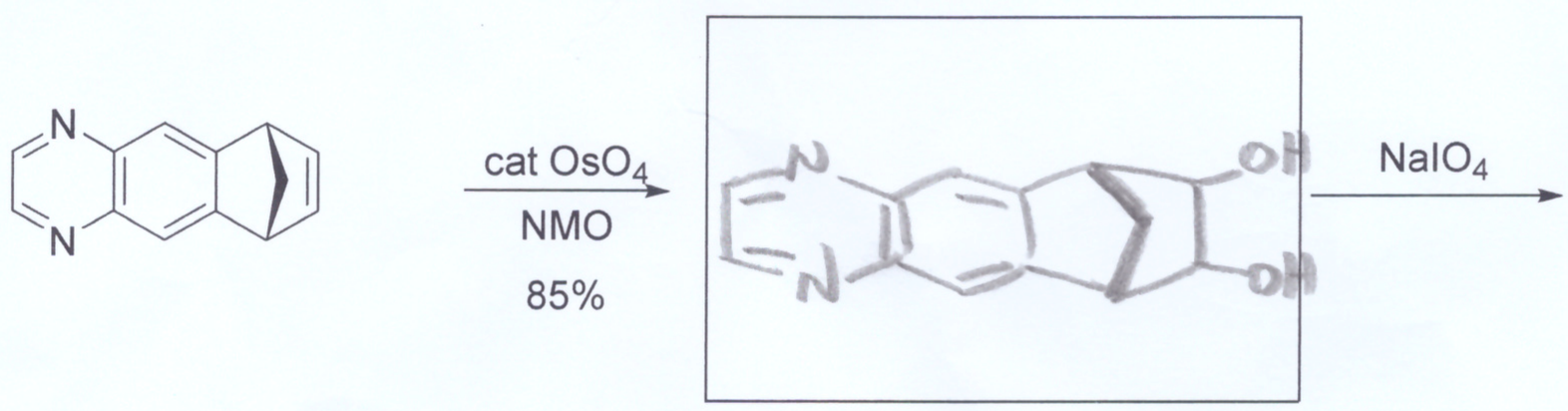
(1)

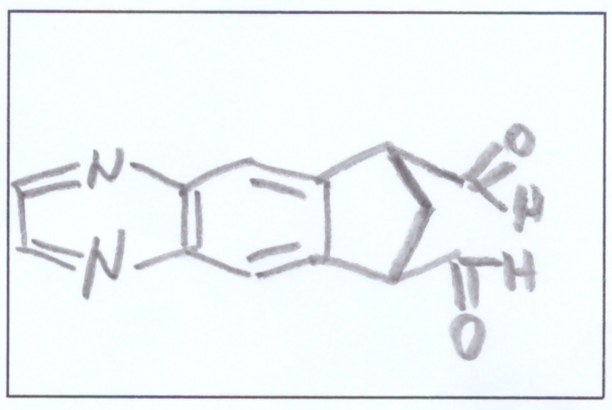


(2)

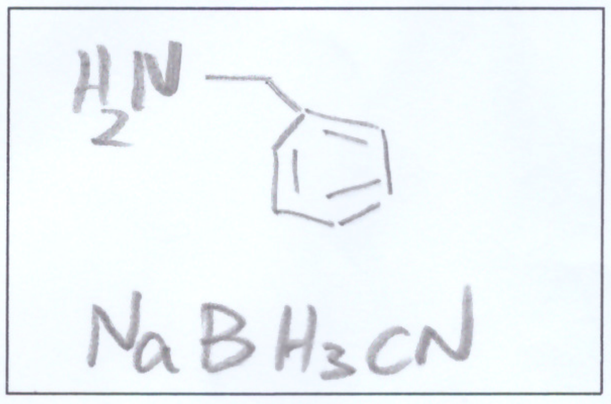


3. Complete the following transformations.



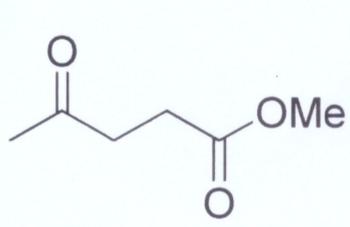


Reductive amination

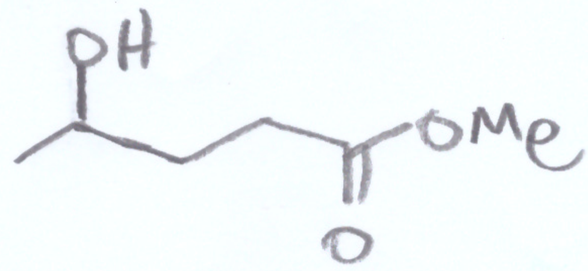


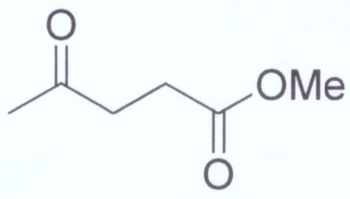
for Varenicline synthesis

4. Predict the reaction products.

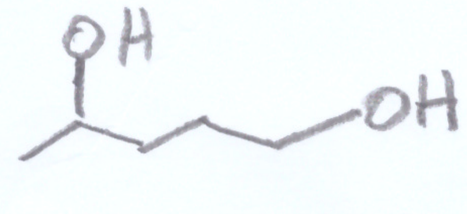


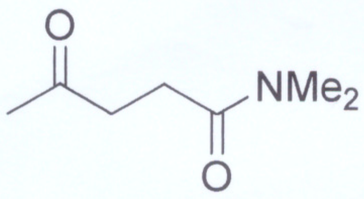
$NaBH_4$



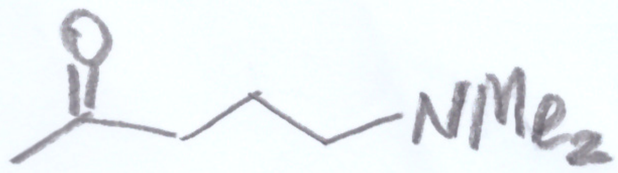


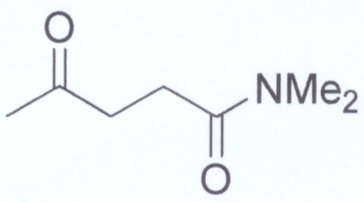
$LiAlH_4$



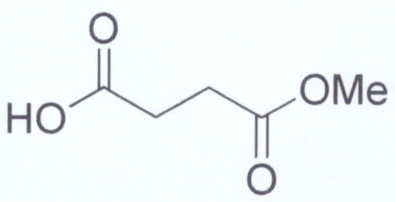


BH_3

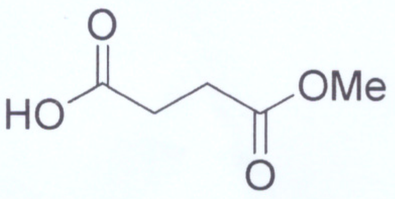




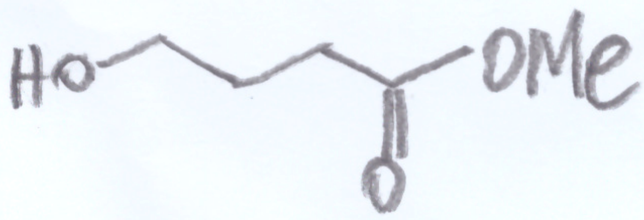
$LiAlH_4$



$LiBH_4$



BH_3



5. Proposed a synthetic route for the following transformation using the types of reactions that we have covered in the lectures.